



Course Name

GENERAL PRACTICAL PHYSICS II

Topic: Introduction to Ohmic Resistance and
Experiment to Verify Ohm's Law

WEEK 2

IN SUMMARY

This week, I learned about Ohmic resistance, a fundamental concept in physics. Ohmic resistance is a property of materials that exhibit a constant resistance over a wide range of current and voltage values. This means that the relationship between voltage and current for an Ohmic resistor is a straight line, which is described by Ohm's Law. I also learned about the different types of resistors, including Ohmic and non-Ohmic resistors. Non-Ohmic resistors have a resistance that changes with voltage or current. Some examples of non-Ohmic resistors include varistors, photoresistors, and thermistors.

BULLET POINT SUMMARY

- Ohm's Law states that the current passing through a conductor is directly proportional to the voltage across it, provided the temperature remains constant.
- Ohmic resistance is a property of a material that exhibits a constant resistance over a wide range of current and voltage values.
- The connection between Ohm's Law and Ohmic resistance lies in the fact that Ohm's Law provides an equation that explains the behaviour of materials or components exhibiting ohmic resistance.
- ◊ Ohmic resistors have the following characteristics:
 - ◊ Their resistance remains constant regardless of the voltage applied or the current flowing through them.
- Their behaviour is reversible, meaning that if the current direction is reversed, the voltage across the resistor also reverses.



CASE STUDY

Title: Using Ohmic Resistance to Control the Brightness of a Light Bulb

Introduction: Ohmic resistance can be used to control the brightness of a light bulb by varying the amount of current that flows through it. This is because the brightness of a light bulb is directly proportional to the amount of current that flows through it.

Objective: The objective of this life case study is to design and build a circuit to control the brightness of a light bulb using Ohmic resistance.

CASE STUDY

Methodology:

Design a circuit that includes a light bulb, a resistor, and a variable resistor.

Build the circuit and connect it to a power source.

Vary the resistance of the variable resistor and observe the brightness of the light bulb.

Expected Results:

As the resistance of the variable resistor is increased, the brightness of the light bulb will decrease.

This is because the increased resistance will reduce the amount of current that flows through the light bulb.

Conclusion:

This life case study has demonstrated how Ohmic resistance can be used to control the brightness of a light bulb. This principle is used in a variety of real-world applications, such as dimmers for lights and volume controls for audio systems.

QUESTIONS TO PONDER

- What are some factors that can affect the resistance of a material?
- How can Ohmic resistance be used to design and build electrical circuits?
- What are some real-world applications of Ohmic resistance?

SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

Ability to explain the concept of Ohmic resistance and its characteristics.

Ability to apply Ohm's Law to solve circuit problems.

Ability to verify Ohm's Law experimentally.

PERSONAL REFLECTION

I found this week's lesson on Ohmic resistance to be very informative and interesting. I learned a lot about the fundamental concepts of Ohmic resistance and its applications in the real world.

